



A terahertz metamaterial based on electromagnetically induced transparency effect and its sensing performance

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ARTICLE INFO

Keywords:

Terahertz
Electromagnetically induced transparency (EIT)
Metamaterial
Sensor

ABSTRACT

We present a terahertz metamaterial based on electromagnetically induced transparency (EIT), of which the unit cell is made up of the coupled “bright” circular split-ring resonator (CSRR) and “dark” square split-ring resonator (SSRR) with practically equal resonance frequency. With the strong coupling of bright mode and dark mode, a sharply narrow transparency peak is observed at terahertz region. Then, the influences of the physical parameters on EIT-like effect are simulated and analyzed by the electromagnetic simulation software CST. Furthermore it is numerically demonstrated that the EIT-like metamaterial is a promising candidate for sensing with refractive index sensitivity of 96.2 GHz/RIU, which means that the transmission peak of the sensor shifts 96.2 GHz per unit change of refractive index of the surrounding medium. Finally, transmission responses of the sensor based on EIT-like effect are investigated by terahertz time-domain spectroscopy, showing a good perception capability consistent with the simulation results. With the property of high refractive index sensitivity, the metamaterial can play an important role in terahertz sensing and detection technology.

1. Introduction

Electromagnetically induced transparency (EIT), a quantum interference effect that occurs in a laser-driven atomic system, can lead to an extremely narrowband transparency window over a wide absorption spectrum [1–3]. In general, the EIT-like spectral response is a result of bright–bright mode coupling [4] or bright–dark mode coupling [5,6]. The resonance of the bright element owns a low quality factor. Meanwhile, the dark element usually supports a sharper resonance profile and hardly interacts with the incident waves. The EIT-like spectral response of the metamaterial has been verified in the microwave [7,8], terahertz [9,10] and optical bands [11,12], which provides a new approach of controlling electromagnetic waves. Characterized by the narrow transmission window and a strong field confinement, the EIT-like metamaterial has attracted an increasing attention in many aspects, such as modulators [13,14], low-loss slow-light devices [11,15], and especially in the sensing applications [16,17]. In 2007, a metamaterial resonator consisting of split ring resonators (SRRs) was proposed by Driscoll et al. [18]. The resonator can be used in the detection of biochemical substances. Due to the LC resonance and low Q-value, the resonator possesses low sensing sensitivity. In order to improve the sensing performance, a great number of metamaterial sensors based on EIT-like effect have been proposed. In 2009, Chen et al. [19] proposed two SRRs based on asymmetrically coupled resonator (ACR) in optical range, the simulation results show a high quality factor that highly required for label-free bio-sensors. In 2012, a metamaterial based on

EIT-like effect is designed by Meng et al. [20], which is applied to the refractive index sensing. Moreover, in 2013, He et al. [21] proposed a new idea and used Babinet principle to design a complementary metamaterial. The structure unit is composed of two complementary nanoscale metal film in a uniform surface. The simulation result shows that the resonator achieves the refractive index sensing with sensitivity of 928.9 nm/RIU. Because of common drawbacks of the low sensitivity and low Q-value, most reported sensors cannot satisfy the practical applications. Therefore, the design of a sensor with high sensitivity shows important research and application value.

In this paper, a metamaterial based on EIT-like effect is designed. The unit of the metamaterial structure comprises coupled “bright” CSRR and “dark” SSRR. The result exhibits a sharply narrow transparency peak in transmission spectrum at 0.683 THz. Then, the mechanism of the EIT-like is illustrated by investigating the surface current distributions. Besides, the influences of the physical parameters on EIT-like effect have been analyzed by commercial finite integration package (CST Microwave Studio). Finally, the refractive index sensing of this metamaterial with refractive sensitivity of 96.2 and FOM of 7.6 has been simulated and validated, which are better than the reference [2] and [5].

2. Theoretical analysis of EIT effect

The concept of electromagnetic induced transparency (EIT) involves a coherent process derived from the media transparent phenomenon

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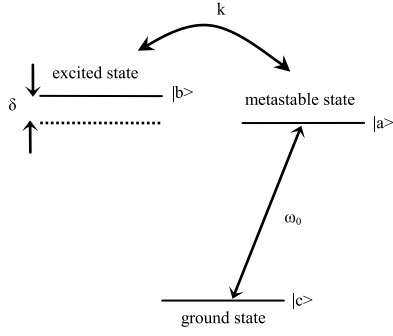


Fig. 1. V type three level system energy diagram.

that occurs in the three-level atomic systems [22]. A v type three-level atom system for classical EIT effect is shown in Fig. 1. $|c\rangle$ is the ground state, $|b\rangle$ is the excited state and $|a\rangle$ is metastable state. $|c\rangle \rightarrow |a\rangle$ is defined as even level of allowed transition. $|c\rangle \rightarrow |b\rangle$ is defined as the excitation of the dipole forbidden transitions, which will appear in the couple of quadrupole dipole mode (dark mode) and dipole mode (bright mode). δ indicates the detuning of the transition between the excited state and metastable state. The transition probability k between state $|a\rangle$ and $|b\rangle$ is related to the degree of coupling of quadrupole dipole and dipole. After the excitation of pump light, the two possible pathways are $|c\rangle \rightarrow |a\rangle$ and $|c\rangle \rightarrow |a\rangle \rightarrow |b\rangle \rightarrow |a\rangle$. The destructive interfere happens, which brings a transmittance peak at a wide absorption spectrum [23]. The bright mode with a larger radiation loss and lower quality factor can directly couple with the incident light. The dark mode with a larger quality can only couple with the bright mode, which is only subject to the limitation of the natural loss of the material. Because of the strong coupling between the two modes, the coupling resonator (CR) will produce destructive interference and appear EIT-like effect with a sharp transparent window.

To provide a clearer description of the EIT-like effect, two simple oscillator models have been availably demonstrated in [24–26]. The bright mode is expressed by the oscillator q_1 , which is driven by the source $E_0(t)$ ($E_0(t) = E_0 e^{i\omega t}$). The oscillator q_2 represents a dark mode that can only be excited by the coupling between the two oscillators. The coupling of $q_1(t)$ and $q_2(t)$ satisfies the following equations:

$$\ddot{q}_1(t) + \gamma_1 \dot{q}_1(t) + \omega_0^2 q_1(t) + k q_2(t) = g E_0(t) \quad (1)$$

$$\ddot{q}_2(t) + \gamma_2 \dot{q}_2(t) + (\omega_0 + \delta) q_2(t) + k q_1(t) = 0 \quad (2)$$

The (γ_1, γ_2) ($\gamma_2 \ll \gamma_1 \ll \omega_0$) and (q_1, q_2) are the damping and the resonant amplitude of oscillator q_1 and q_2 , respectively. ω_0 is the resonance frequency that q_1 isolates from q_2 . g represents the coupling strength between bright mode oscillator and the external field. The δ and k represent the frequency detuning factor and coupling coefficient between q_1 and q_2 . Under the condition of $\omega - \omega_0 \ll \omega_0$ and $\omega_0^2 - \omega^2 \approx -2\omega_0(\omega - \omega_0)$, we can obtain the transmission as follow [26]:

$$T = 1 - \text{Re} \frac{i g^2 (\omega - \omega_0 - \delta + i \gamma_2 / 2)}{(\omega - \omega_0 + i \gamma_1 / 2)(\omega - \omega_0 - \delta + i \gamma_2 / 2) - \Omega^2 / 4} \quad (3)$$

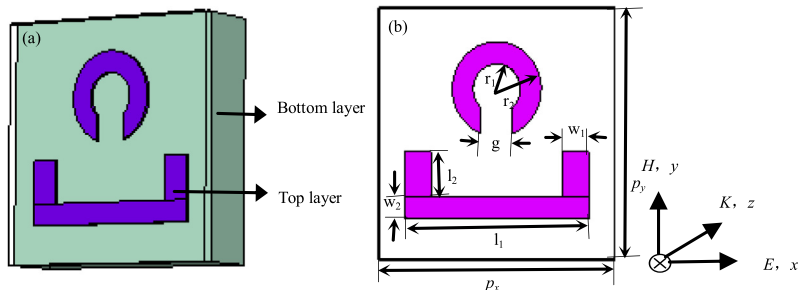


Fig. 2. Schematic diagram of the proposed metamaterial resonator unit cell.

In this paper, we use the coupling of the bright mode (CSRR) and the dark mode (SSRR) to obtain EIT-like effect. When the detuning δ is zero, there is a narrow transmission peak because of the increase of the coupling detuning.

3. Structure design and simulation results

The unit cell of the proposed metamaterial is a classic dual-layer structure composed of a dielectric layer and a metallic layer. As is illustrated in Fig. 2, the bottom layer is a dielectric layer with thickness of $100 \mu\text{m}$ and the top layer is a metal layer that consists of a circular split-ring and a square split-ring. The material of dielectric layer is typical loss quartz crystal with a dielectric constant of $\epsilon = 3.75 + i0.2$. Both the circular split-ring and square split-ring are made of copper with a conductivity of $5.96 \times 10^7 \text{ S/m}$. The thickness of those two split-ring are $5 \mu\text{m}$. The period of the unit $P_x = P_y = 180 \mu\text{m}$, the inner radius of circular rings $r_1 = 18 \mu\text{m}$, the outer radius $r_2 = 34 \mu\text{m}$, the width of SSRR $w_1 = 20 \mu\text{m}$, $w_2 = 16 \mu\text{m}$, the length of SSRR $l_1 = 140 \mu\text{m}$, $l_2 = 32 \mu\text{m}$, and the opening size of CSRR $g = 24 \mu\text{m}$. The metamaterial resonator is simulated and optimized by CST Microwave Studio 2013 based on the finite difference time domain (FDTD) solver. In the simulation procedure, we set the electric boundary along the x -direction, magnetic boundary along the y -direction and open (add space) boundary along the z -direction respectively. The propagation direction of electromagnetic wave is perpendicular to the plane of the structure; the electric field polarization is in x -direction.

As shown in Fig. 3, the simulated transmission curve of the resonator is depicted, as well as the results for CSRR and SSRR. The resonant point for CSRR and SSRR alone is nearly equal resonance frequency at 0.688 THz and 0.693 THz , the Full Width at Half Maximum (FWHM) and quality factor value (Q-value) are quite different. It can be found that the CSRR alone with large Q-value, which is regarded as a bright mode because of the strong coupling between the CSRR and the incident wave. In contrast, the SSRR alone has weak interaction with the incident wave, making it known as the dark mode with a low Q-value. When combining the CSRR with SSRR, an additional transmission peak with higher Q-value emerges at 0.683 THz (ω_2) and the two transmission dips appear at 0.662 THz and 0.692 THz (ω_1, ω_3) due to the destructive interference. Such a transparency window produced by the coupling of the dark and bright element is commonly regarded as metamaterial-based analog of the EIT effect.

In order to analyze the physical mechanism of the EIT-like effect, the current distributions of the resonator on the two resonant dips at ω_1 and ω_3 and the transmission peak at ω_2 are depicted in Fig. 4(a), (b), (c). The surface current distribution of ω_1 presents the clockwise induced currents in CSRR and SSRR, which is similar to the inductive capacitive (LC) resonance in nature. The surface current distribution at ω_3 possesses the anticlockwise surface currents, which stands for the dipole resonance that strongly coupling to free space. Thus, the currents are weaker than those at ω_1 . At the EIT-like peak ω_2 , the current direction of CSRR is parallel to the positive direction of the x -axis. However, the current direction of SSRR is parallel to the negative direction of the x -axis. The currents at peak ω_2 are stronger than those at the other two transmission dips ω_1, ω_3 owing to the destructive interface between the two resonances at ω_1 and ω_3 .

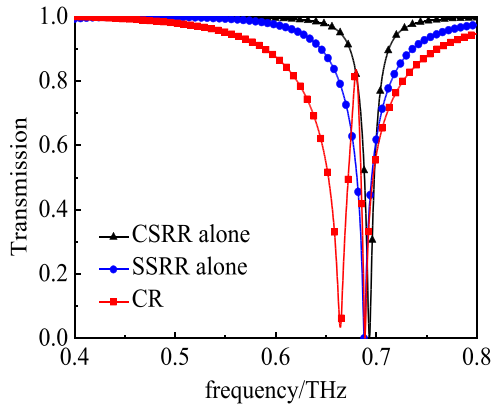


Fig. 3. Simulated transmission curve of CSRR alone (black), SSRR alone (green), and CR (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. The influence of structural parameters on the EIT-like effect

Most of the metamaterial are highly sensitive to polarization state of incident waves. So we next investigate the influence of different incident waves on the sensing performance. As we can see from Fig. 5(a), the transmission curve of TE wave shows a great agreement with the one of TM wave, indicating the metamaterial is polarization-independent. Compared with the polarization sensitive metamaterials, polarization independent ones are more convenient for practical use because they can endure changes in polarization states of the incident wave.

Besides, the influences of the physical parameters on EIT-like effects of the metamaterial are analyzed. Fig. 5(b) and (c) display the transmission spectra with the different g and the different distance d between CSRR and SSRR respectively. The coupling strength between the bright mode and the dark mode become stronger when g is from $18 \mu\text{m}$ to $24 \mu\text{m}$. The EIT-like effect is more obvious, FWHM becomes narrower, leading a higher Q-value. For example, we can obtain a small Q of 14 when the g is $18 \mu\text{m}$. In contrast, a large Q of 49 is obtained when a larger opening size g is $24 \mu\text{m}$. What is more, the EIT-like effect and FWHM were closely related to the distance of CSRR and SSRR d . The FWHM decreased with the increase of d . The Q-value of the metamaterial becomes larger. We can get the optimal Q with 63 when $g = 24 \mu\text{m}$ and $d = 85 \mu\text{m}$.

Finally, we investigate the impact of the dielectric losses on the performance of the metamaterial. Fig. 5(d) shows the transmission when the $\tan\delta = 0.004, 0.0004, 0.00004$ and dielectric layer loss free. With the bigger dielectric losses, the EIT-like effect becomes weaker, FWHM becomes wider, while the resonant frequency is almost the same, leading to a smaller Q-value.

5. Refractive-index-based sensor

It can be seen that the EIT-like narrow window and strong electromagnetic enhancement appear at resonance point where the dark SSRR

Table 1

Comparison of the proposed structure with other reported sensors.

Sensor	Wave band	Couple mode	Sensitivity	FOM
Yahiaoui R et al. [10]	0.15~0.85 THz	Plasmon	54.18 GHz/RIU	~
Chen et al. [19]	0~100.00 THz	EIT-like	1200.00 nm/U	2.86
Meng et al. [20]	1.20~2.00 GHz	EIT-like	77.25 nm/RIU	8.14
Longqing Cong et al. [29]	0.50~0.90 THz	Plasmon	~	2.30
Proposed structure	0.40~0.80 THz	EIT-like	96.2 GHz/RIU	7.80

and the bright CSRR couple. The metamaterial can behave as a resonant cavity and its resonant frequency depends on the dielectric condition of surrounding media. Therefore, the resonator based on EIT-like effect can be considered as a distinctly promising candidate for refractive index sensors [27]. We evaluate the sensing performance of the CR by measuring the transmission spectral shift of the resonator. In simulation, it is needed to add a analyte layer with thickness of $5 \mu\text{m}$ on the resonator. Fig. 6 illustrates the simulated transmission curve of sensing. A red-shift of the transmission peak is obvious when we increase the refractive index of the analyzed medium from 1.0 to 2.5. The refractive index sensitivity (S) of the sensor is one of the most important indicators of the sensing performance and is defined as the variation of the resonant frequency of the unit refractive index. The definition of S is $S = \Delta f / \Delta n$ and the measurement unit is GHz/RIU, where Δf is the variation of the resonant frequency and Δn is the change of refractive index. We get a highly refractive index sensitivity of CR with 96.2 GHz/RIU.

In addition, the figure of merit (FOM) [28] is another key factor to evaluate the refractive index sensor characteristics considering the influence of bandwidth on the sensing performance, which is defined as:

$$\text{FOM} = \frac{S \text{ (GHz/RIU)}}{\text{FWHM (GHz)}} \quad (4)$$

where S is sensitivity, FWHM represents the half width of the transmission peak. We figure out the FOM values of the CSRR, SRR and CR are 4.9, 5.7, 7.8 respectively. Because resonant peak of CR has a narrower bandwidth, its FOM value shows significantly higher than other two individual SRRs. Furthermore, the sensitivity and FOM of different sensors which were mentioned previously have been compared with our presented structure in Table 1. It is found that the proposed sensor reaches a relatively high sensitivity and FOM as compared to [10,19,20,29].

6. Experiment demonstration

The metamaterial of $5 \mu\text{m}$ -thick copper is fabricated on a silicon wafer with the thick of $100 \mu\text{m}$ using conventional optical lithography technology. The optical microscope image of the fabricated sensor sample is shown in Fig. 7(a), which has 50 unit cells along x and y axis, respectively. Therefore, the physical dimensions of the sensor is $9 \text{ mm} \times 9 \text{ mm}$. Then, in order to demonstrate the sensing performance, we use a customized circular complementary double-sided adhesive to seal water–alcohol mixture of different concentrations between sensor and cover glass with 0.5 mm . The water–alcohol mixtures were prepared

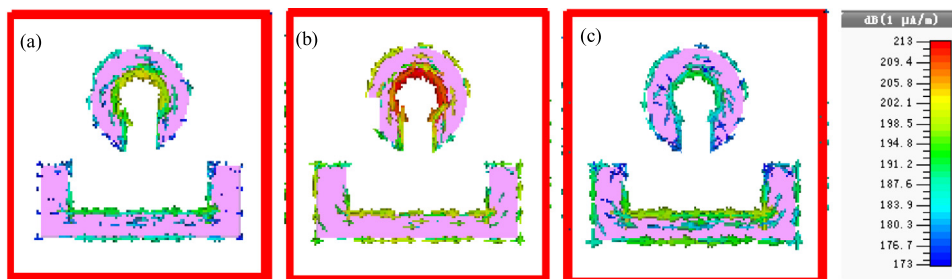


Fig. 4. The surface current distributions at (a) 0.622 THz (ω_1), (b) 0.683 THz (ω_2), and (c) 0.692 THz (ω_3).

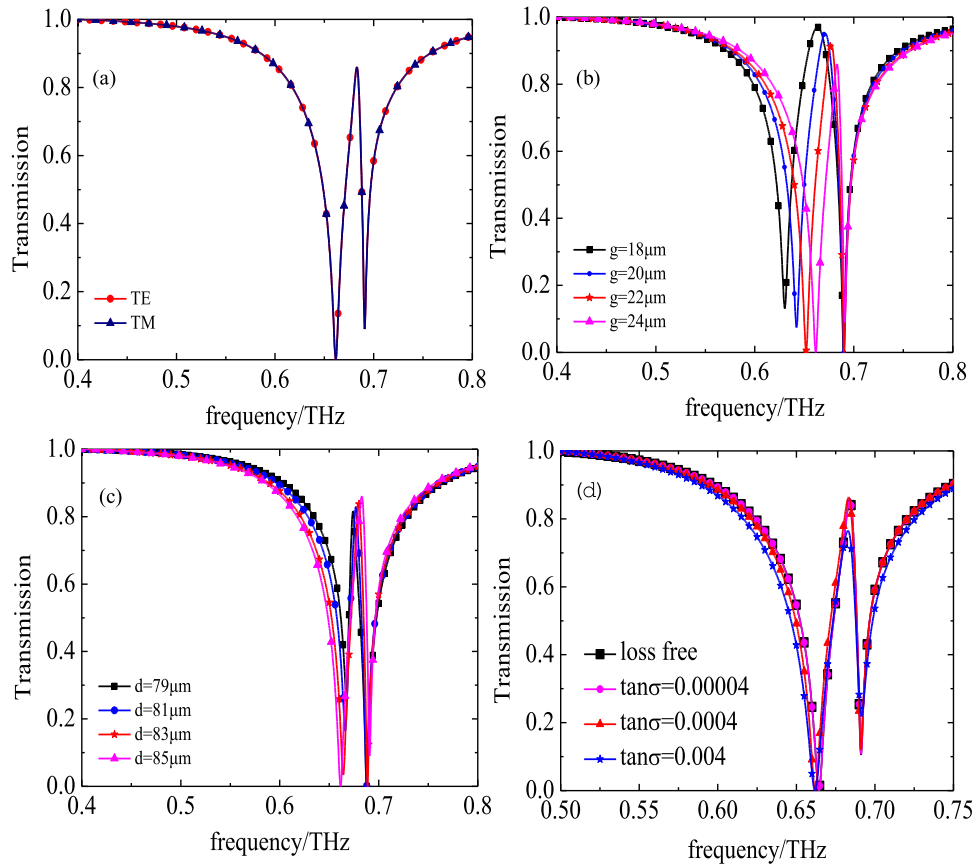


Fig. 5. Simulated transmission spectra for TE and TM polarization (a), on different opening size g (b), distance d (c) with CSRR and SSRR, and on different loss tangent (d) with the dielectric.

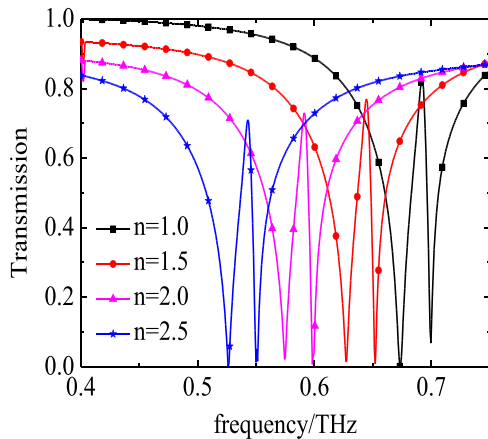


Fig. 6. Simulated resonance shift of the EIT-like metamaterial with respect to a change of the refractive index of the surrounding medium.

with 100%, 70% and 40% concentration of alcohol, respectively. The transmission of sealed sensor and cover glass filled with no solution was set as a reference.

Fig. 7(b), (c) show the comparison of measured and simulated transmission spectra with mixture of different concentrations. Due to the refractive index around the metamaterial, the red-shift of the EIT peak occurs when the volume fraction of water increases. At the same time, the amplitude of the transmission peak decrease, which results in less absorption of the EIT transmission dips. Besides, it is noted that the EIT-like effect becomes weaker, which comes from the weakly coupling

between the bright mode and dark mode with the increasing in the volume fraction of water.

As shown in Fig. 7(d), the red-shift of frequencies at the EIT-like peak and transmission dips almost linearly decreases with the decrease of alcohol concentration. We can clearly see that the maximum red-shift is about 112.0 GHz when concentration of the mixture varies from 100% to 40%. The above experiment results further prove that the good sensing performance of metamaterial based on EIT-like effect has been obtained.

7. Conclusion

In conclusion, we presented a refractive index sensor. Simulation results show that the metamaterial achieves EIT-like effect and creates a sharp transmission peak. The proposed metamaterial is characteristic of polarization insensitivity, which is more convenient for practical use because of great endurance of changes in polarization states of incident waves. Then, the sensor realizes a refractive index sensing with sensitivity of 96.2 GHz/RIU, FOM of 7.8. Besides, we demonstrate the sensing performance with our experiment, it realizes the maximum frequency shift with 112.0 GHz. The proposed metamaterial refractive index sensor based on EIT-like in the terahertz range has many potential applications in environmental, chemical and biological diagnosis.

Acknowledgments

This work was supported by New Direction Cultivation Project of Chongqing University of Posts and Telecommunications, China (A2014-116), the Key Research Program of Chongqing University of Posts and Telecommunications, China on Interdisciplinary and emerging field (A2018-01).

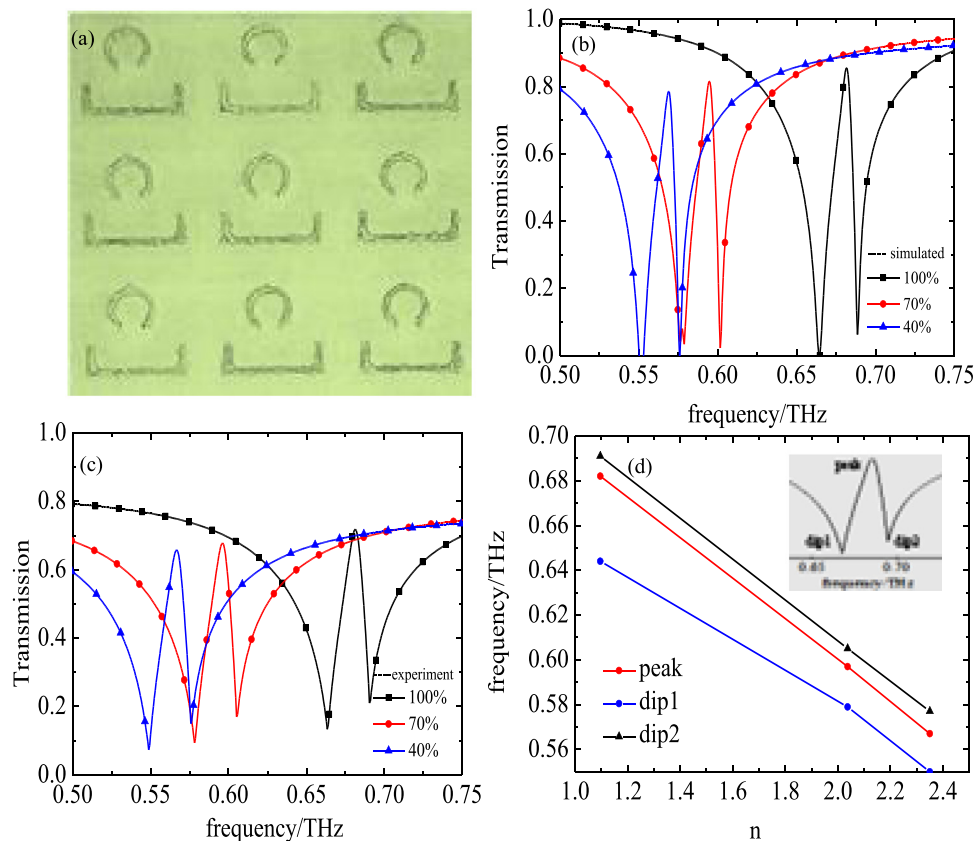


Fig. 7. (a) Optical microscopic image of the fabricated sample. (b) (c) the simulated and measured results of the EIT-based sensor in different water–alcohol mixture. (d) the experimental red-shift in frequencies at the EIT peak and transmission dips.

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